

Roll No.

26/8148**B.C.A. (Fifth Semester)****Examination, 2026****First Paper****(Introduction to DBMS)***Time : Two Hours]**[Maximum Marks : 75***Note :** Attempt **all** sections as directed.**Section-A****(Very Short Answer Type Questions)**

Note : Attempt any **5 (five)** questions. Each question carries **2 (two)** marks and the answer of each question should not exceed **50** Words. $5 \times 2 = 10$

1. ~~(a)~~ Demonstrate Functional dependency
(b) what is log files?

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- ~~(c)~~ What are the responsibilities of a DBA?
- ~~(d)~~ What is the difference between the strong entity set and weak entity set?
- ~~(e)~~ Explain the concept of Recovery.

Section-B**(Short Answer Type Questions)**

Note : Attempt any **5 (five)** out of total **8 (eight)** questions. Each questions carries **5 (five)** marks and the of each question should not exceed **100** words.

 $5 \times 5 = 25$

2. ~~(a)~~ Define data model. List different types of data models.
~~(b)~~ What is data independence? What are the differences between logical data Independence and physical data independence?
~~(c)~~ What is transaction? Explain the ACID Properties of transactions.

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- (d) What is referential integrity Constraints? Give a simple example.
- (e) What is Entity-Relationship model? Give symbol used in E-R Diagram.
- (f) How does a check point mechanism help in database recovery?
- (g) What is deadlock? When it occurs?
- (h) Discuss the different types of keys in DBMS: super key, candidate key, primary. Provide examples for each.

Section-C

(Long Answer Type Questions)

Note : Attempt any **2 (two)** out of **4 (four)** questions. Each questions carries **20 (twenty)** marks and the answer of each question should not exceed **400** words.

$$2 \times 20 = 40$$

3. (a) Explain ANSI-SPARC 3Level architecture of DBMS and Give the suitable diagram in support of you explanation.

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- (b) Discuss the Relational Algebra in depth. Explain the fundamental operators of relational Algebra with the help of examples.
- (c) What is normalization? What are the conditions required for a relation to be in 2NF and 3NF explain with examples.
- (d) Discuss Data definition Language (DDL) and Data Manipulation Language (DML) in SQL. Write advanced examples of CREATE TABLE, ALTER TABLE, DROP, TRUNCATE, INSERT INTO SELECT, DELETE with subqueries.